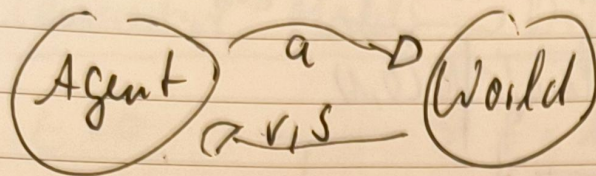


# Intro Reinforcement Learning



$$\text{Max } E \left[ \sum_t \gamma^t r_t \right]$$

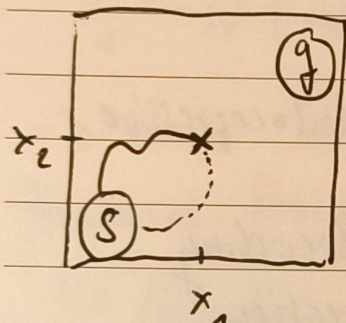
$$\gamma \in [0, 1]$$

"Max sum of future rewards"

State: sufficient statistic of history to predict the future

History:  $h = \{a_0, s_0, r_0, a_1, s_1, r_1, \dots, r_t\}$

$\uparrow$  action       $\uparrow$  reward



location starts  $s$ , wants to goal  $g$  and is at state  $(x_1, x_2)$  "location".

$$h = a_0, s_0, r_0, a_1, s_1, r_1, \dots, s_t$$

policy:  $\pi(s)$  or  $\pi(s, a) = \Pr(a) \in s$

$\pi^*$  is the optimal policy

reward:  $R(s, a)$ : reward for doing  $a$  in  $s$

discount factor:  $\gamma$       from world:  $T$  = Transition Function

$$T(s' | s, a) : \text{Pr of } s' \text{ if do } a \text{ in } s$$

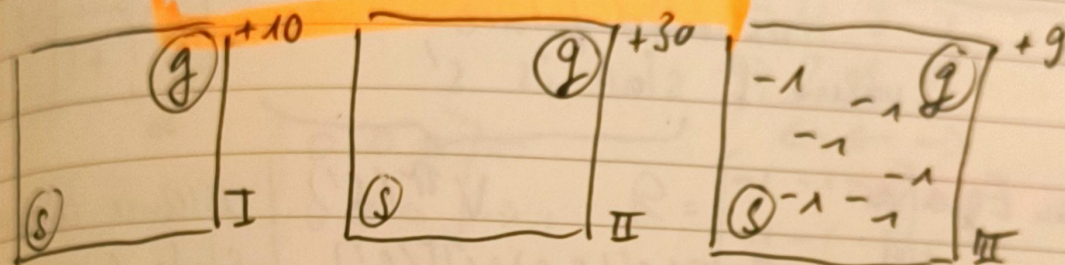
$\Rightarrow$  Markov Decision Process:  $\{S, A, R, T, \gamma, \}$

Planning we have MDP available

$\uparrow$  start  $\mu$



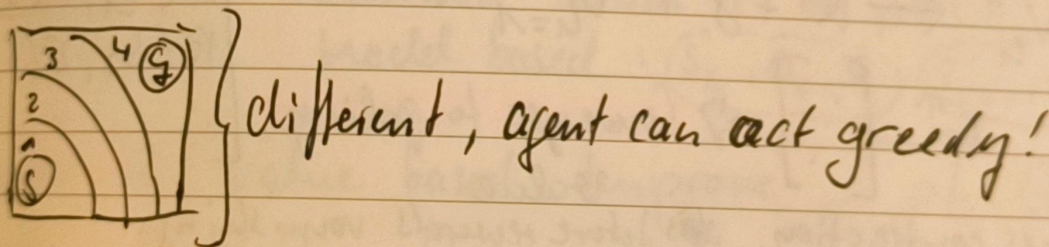
Goal:  $\max_{\pi} \mathbb{E}_{\pi, T} \left[ \sum_{t=0}^{\infty} \gamma^t r_t \right]$



I vs II:  $\mathbb{E}[\sum_{t=0}^{\infty} \gamma^t C_1 r_t] = C_1 \cdot \mathbb{E}[\sum_{t=0}^{\infty} \gamma^t r_t]$ : same

vs III: same, just adding constant (-1), but not influencing reward function!

$\Rightarrow$  all same

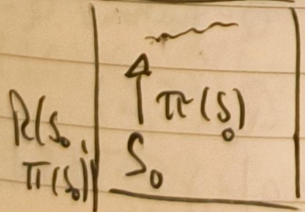


Policy Evaluation for given  $\pi$ : Value of  $\pi$  at  $s$

$$V^{\pi}(s) \equiv \mathbb{E}_{\pi, T} \left[ \sum_{t=0}^{\infty} \gamma^t r_t \mid S_0 = s \right]$$

$$V^{\pi}(s) = \mathbb{E} \left[ \cancel{\gamma V_0} + \sum_{t=1}^{\infty} \gamma^t r_t \mid S_0 = s \right] \quad \text{assume } \pi(s) \text{ is deterministic}$$

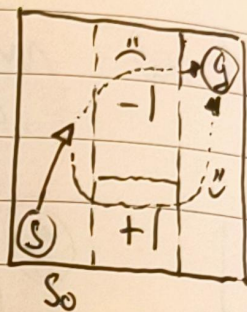
$$\underbrace{R(s, \pi(s))}_{\text{current reward}} + \mathbb{E} \left[ \sum_{t=1}^{\infty} \gamma^t r_t \mid S_0 = s \right]$$



$$\mathbb{E} \left[ \gamma \sum_{s'} \sum_{t=0}^{\infty} \gamma^t T(s' | s_0, \pi | s_0) r_t \mid s_0 = s \right]$$



$$\underbrace{\gamma}_{\text{discount}} \sum_{s'} T(s'|s, \pi(s)) \underbrace{E \left[ \sum_{t=0}^{\infty} \gamma^t R_t \mid S_0 = s' \right]}_{\text{expected future value if start is } s'}$$



**Bellman Equation:**

$$V^{\pi}(s) = R(s, \pi(s)) + \gamma \sum_{s'} T(s'|s, \pi(s)) V^{\pi}(s') \quad \left| \begin{array}{l} V^{\pi}(s') \\ \text{assume discrete} \\ \text{state space} \end{array} \right.$$

if  $s$  is discrete:  $V$  is vector, suppose:  $\begin{bmatrix} V^{\pi}(s_1) \\ V^{\pi}(s_2) \\ \vdots \end{bmatrix}$   
 Set  $V_{k=0}^{\pi}$  to arbitrary value.

$$V_k^{\pi} \leftarrow R^{\pi} + \gamma T^{\pi} V_{k-1}^{\pi} \quad \text{with } R^{\pi} = \begin{bmatrix} R(s_1, \pi(s_1)) \\ R(s_2, \pi(s_2)) \\ \vdots \end{bmatrix}$$

and  $T^{\pi}: \begin{bmatrix} \vdots \end{bmatrix} \Rightarrow$  **converges to policy**  $\left[ \begin{array}{l} \vdots \\ \text{value} \end{array} \right]$

$\gamma$  implies contraction, ~~dis~~ future rewards vanishing.

$$V_{k=1}^{\pi} \leftarrow R^{\pi} + \gamma T^{\pi} V_0^{\pi}; \quad V_{k=2}^{\pi} \leftarrow R^{\pi} + \gamma T^{\pi} V_{k=1}^{\pi} =$$

$$\text{Opt: find best } \pi \quad \left\| \begin{array}{l} = R^{\pi} + \gamma T^{\pi} (R^{\pi} + \gamma T^{\pi} V_0^{\pi}) = \\ = R^{\pi} + \gamma T^{\pi} R^{\pi} + \gamma^2 T^{\pi} T^{\pi} V_0^{\pi} \\ \underbrace{\hspace{1cm}}_{\text{now}} \quad \underbrace{\hspace{1cm}}_{\text{1 step fut.}} \quad \underbrace{\hspace{1cm}}_{\text{2 steps fut.}} \end{array} \right.$$

$$Q^{\pi}(s, a) = R(s, a) + \gamma \sum_{s'} T(s'|s, a) \cdot V^{\pi}(s')$$

**Policy Improvement**  $\pi(s) \leftarrow \operatorname{argmax}_a Q^{\pi}(s, a)$

Approach #1:

1. arb.  $\pi$
2. eval  $\pi$
3. improve  $\pi$

Approach #2: Q-Function / Q-opt

1. Start arbit.  $Q_0(s, a)$
- $Q_k(s, a) \leftarrow \underbrace{R(s, a)}_{\text{eval}} + \gamma \sum_{s'} T(s'|s, a) Q_{k-1}(s')$



e.g.  $\boxed{+5 \mid -1 \mid -1 \mid -1 \mid +10}$

$\pi$  is fixed mapping of state space

#1 end  $\leftarrow \leftarrow \leftarrow$  end

V: 5 4 3 2 10, Q:  $\leftarrow 4 \ 3 \ 2$   
 $\rightarrow 2 \ 1 \ 9$

#2  $Q_0(s,a) = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \leftarrow \begin{matrix} \leftarrow \leftarrow \rightarrow \\ \leftarrow \leftarrow \rightarrow \end{matrix}$ ,  $R = \begin{bmatrix} -1 & -1 & -1 \\ -1 & -1 & -1 \end{bmatrix}$

$$Q_0(s,a) \leftarrow \begin{bmatrix} -1 & -1 & -1 \\ -1 & -1 & -1 \end{bmatrix} + \gamma \begin{bmatrix} 5 & 0 & 0 \\ 0 & 0 & 10 \end{bmatrix} = Q_1 = \begin{bmatrix} 4 & -1 & -1 \\ -1 & -1 & 10 \end{bmatrix}$$

RL: We don't have T (Transition Function), but h!

option 1: We can learn T or  $p(T)$

Model based:  $(\underbrace{s_t, a_t, r_t}_{\text{Model}}, \underbrace{s_{t+1}, a_{t+1}, r_{t+1}}_{\text{Model}})$

Option 2: Value based: Improve ... of Q

Option 3: Policy based: Ways to compute  $V$  of  $V^\pi$  w.r.t.  $\pi$  (gradient)

Acc. to Opt. 2: A: how to choose an action?

B: how to adj. Q given results of actions?

A: Explore & Exploit trade off:  $\epsilon$ -greedy,

B:  $\max Q$  w.r.t.  $1-\epsilon$  (old est. val.), random w.r.t.  $\epsilon$  (old)

B:  $Q(s,a) \leftarrow \widehat{Q}(s,a) + \alpha_t [r + \gamma \widehat{Q}(s',a') - \widehat{Q}(s,a)]$

A:  $Q(s,a) \leftarrow \widehat{Q}(s,a) + \alpha_t [r + \gamma \max_{a'} (\widehat{Q}(s',a') - \widehat{Q}(s,a))]$